

KEERTHAN MAHESH

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WORK EXPERIENCE

KonfigAI

Backend Lead

Charlotte, NC

May 2023 - Present

- **Sole architect of the main microservice** for **KonfigAI**, a no-code platform, responsible for the architecture and design of a production-level web application; developed **Agentic AI** capabilities enabling users to generate entire applications from a single prompt; a **multi-agent system** pairing an **LLM orchestrator** with **18 specialized sub-agents** for workflow automation.
- Designed the companion **MCP tool server** (FastMCP over SSE) exposing 100+ Konfig operations; with **tenacity-based retries**, **async connection pooling**, per-request header forwarding, and **token-optimized response encoding**; integrated **7 LLM providers**.
- Optimized **JVM performance** using **GC (G1) tuning**, **-Xms/-Xmx heap sizing**, **pause-time targets**, **heap dump analysis**, and **Java 21 features (Records, Virtual Threads)**, reducing application memory limits by 60%; increasing concurrent users by 200%.
- Built a standalone **AI agent** to **review merge requests**, **suggest enhancements**, **remediate SonarQube issues**, **resolve build failures**, and **analyze Trivy vulnerability scan results**, improving code quality and security review efficiency.
- Developed high-performance, responsive UI using **React.js** and **React Redux**, implementing a **component-driven architecture** using **custom hooks** and efficient **state management**. Optimized application delivery through **code-splitting** and **lazy loading**.

Spatial Modeling and Remote Sensing Lab, University of Maryland

College Park, MD

Research Assistant - Advisor: **Dr. Varaprasad Bandaru**

Aug 2022 - May 2023

- **Nitrogen Recommendation Tool** and **SMaRS Lab**: Spearheaded the development of an end-to-end web application built on **React.js** and **Node.js**, creating an advanced decision support tool for **100+ farmers in the states of Oklahoma and Kansas**.
- Integrated sophisticated **geospatial functionalities** with **React Leaflet**, optimizing location-based data processing and analysis features, with minimal latency; utilized **AWS EC2** for scalable computing and **S3** for secure data storage.

KonfigAI

Software Engineer

Hyderabad, India

Aug 2020 - Aug 2022

- Implemented and optimized scheduled batch operations using **Spring Batch**; processed **200,000 complex transactions in under 5 minutes**, significantly improving system performance and efficiency.
- Designed a multi-layer **caching strategy** using **Redis (LRU cache eviction strategy, custom TTL)** for distributed caching and **Caffeine** for in-memory caching for **connection pool managers** across **MongoDB, MySQL, PostgreSQL, DynamoDB, and SAP HANA**, improving latency, throughput, and backend resource utilization.
- Implemented secure authentication and authorization using **Okta** and **Keycloak**, **JWT token validation**, custom **OAuth 2.0 grant flows - Authorization Code, Implicit, Client Credentials, ROPC - OIDC integration**, **OWASP Top 10 controls** and **role-based access control (RBAC)** to enhance security for user applications.
- Collaborated with **DevSecOps** team using **Jenkins for CI/CD automation**, **Helm charts for Kubernetes deployments**, **Rancher for container orchestration**, **Grafana for log monitoring**, **GC monitoring**, **CPU utilization dashboards**, **OpenTelemetry for observability**, and **OpenBao for secrets management**, improving deployment reliability and operational visibility.
- Used **JProfiler** and **Cryostat** for **JVM profiling**, **runtime diagnostics**, **memory leak analysis**, and **performance monitoring**.
- Engineered comprehensive testing strategy - **unit tests using JUnit and Mockito**, **embedded database tests**, **integration tests**, and **load testing using k6** to validate application reliability, API behavior, and performance under high traffic.

JP Morgan Chase & Co.

Software Development Intern

Hyderabad, India

Jul 2019 - Dec 2019

- Developed a **full-stack progressive web application** using **React.js** and **Golang** that visualizes the relationships between entities in a **Kubernetes** cluster, represented by color to show workload performance.
- With real-time **log streams of pods and container activities** for troubleshooting and monitoring, helped employees **reduce debugging time by 25%**, significantly improving operational efficiency and response times.

PROJECTS

Resume Intelligence: Retrieval-Augmented Generation (RAG) app for Job-Description ↔ Resume matching

- An end-to-end **RAG pipeline**: document ingestion (PDF/DOCX parsing) → **semantic chunking** → **embedding generation** (text-embedding-3-large) → **HNSW-based Approximate Nearest Neighbors (ANN) retrieval** over a **vector store** (Weaviate) → **cross-encoder reranking** → **LLM-driven scoring**, gap analysis, and tailoring recommendations using **FastAPI**.

EDUCATION

University of Maryland, College Park

Master of Engineering in Software Engineering

College Park, MD

Aug 2022 - Dec 2023

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering in Electrical and Electronics Engineering

Goa, India

Aug 2016 - Aug 2020

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, Go, Bash, Assembly, SQL, HTML, CSS, JavaScript, TypeScript, JSON, XML.

Tools and Frameworks: LangGraph, LangChain, MCP, FastAPI, React, React Native, Angular, Spring Boot, Gradle, MongoDB, MySQL, AWS, Kafka, CI/CD, SonarQube, Kubernetes, Docker, Grafana, Jenkins, Spring IoC, Spring JPA, Spring AOP, REST, SOAP.

PUBLICATIONS AND TEACHING

- Agarwal S., Mahesh K., Singh P., Shah J., Sanjeev N. **Investigating Depression Semantics on Reddit using Machine Learning and Deep Learning Techniques.**
- Undergraduate Teaching Assistant for **Object-Oriented Programming in Java - CS F213.**